

Investigation 3.2 - Specific Heat of Pb.

- RESULTS
- Suitable table or similar
 - Units correct
 - Sig-fig correct.

(3)

- CALCULATIONS
- Correct setting out.
 - Correct calculation. (3)
 - Correct units + sig-fig.

DISCUSSION

- Q4 • Comparison to accepted value.
- % difference calculated.

(2)

- Q5 Sources of greatest error • low ΔT value - has largest % uncertainty. (1)

- Q6 Ways to reduce error • Greater amount of Pb
- Use Cu calorimeter in wood - better insulation. (2)

- Q7 • Dry metal - otherwise extra H_2O is added to the system.
- holds extra heat.

(1)

- Q8 • High specific heat $\Rightarrow$ stays hotter for longer - heat is lost slowly. (1)

- Q9 • H_2O is better than metal.
- Has a higher specific heat so stays hotter for longer.

(2)

- CONCLUSION • Statement of value of specific heat. (1)

16

Invest. 4.2 - Heat of Fusion

- RESULTS
- Suitable table or similar
 - Correct units.
 - Correct sig-fig.

(3)

CALCULATIONS

- Correct setting out.
- Correct calculations.
- Correct units + sig-fig. (3)

DISCUSSION

- Q3 • Sources of error (at least 2)
- Estimate of % uncertainty for each.
 - Overall % uncertainty (should add all % uncertainties).

(3)

- Q4. (a) • Impurities - lowers freezing point.
- (b) • Assume melting point is $0^\circ C$ - ΔT will be larger.
- (c) • Value would be lower.

(3)

- Q5 • Above room temp $\Rightarrow$ greater ΔT
- $\Rightarrow$ negates the effect of heat entering from outside the system

} EITHER (1)

$$m_w c_w \Delta T_w = m_i L_f + m_w c_w \Delta T_i$$

- Q6 • States accepted value. (2)
• Calculates % difference.

- Q7 • Changes the E_p of the particles - allows them to move further apart and overcome some attractive forces. (1)

- Q8 • Ice needs to melt first. (2)
• Requires more heat, which is removed from the skin.

CONCLUSION

- Statement of the latent heat of fusion. (1)